HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F78PA

PROBABILITY AND STATISTICS A

IN-PERSON EXAMINATION

December Examination and Assessment Diet 2023

Duration: 2 hours

Total marks available: 60

Answer all 6 questions.
To receive full credit you must show your work and explain your answers, i.e. a correct answer without explanation may yield no credit.

Attached: AMS formula sheet is provided at the start of this exam paper.
Supplied: New Cambridge Statistical Tables
Calculator Permitted: Yes, approved electronic calculators may be used.

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

I understand that infringing this statement would represent an academic misconduct offence subject to disciplinary action according to the University Regulations and Policy regarding Academic Misconduct, with possible significant consequences for degree progression and final degree outcome.

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Some Absolutely Continuous Distributions

Name	Notation	p.d.f.	range	m.g.f.	mean	variance
Uniform	$U(a,b)$	$\frac{1}{b-a}$	$a \le x \le b$	$\frac{e^{b\theta}-e^{a\theta}}{(b-a)\theta}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	$N(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$	$-\infty < x < \infty$	$e^{\mu\theta+\frac{1}{2}\sigma^2\theta^2}$	μ	σ^2
Gamma	$\Gamma(t,\lambda)$	$\frac{\lambda^t}{\Gamma(t)}x^{t-1}e^{-\lambda x}$	$0<x<\infty$, $t,\lambda>0$	$\left(\frac{\lambda}{\lambda-\theta}\right)^t$	$\frac{t}{\lambda}$	$\frac{t}{\lambda^2}$
Exponential	$\text{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$0<x<\infty$, $\lambda>0$	$\frac{\lambda}{\lambda-\theta}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Chi-square	χ^2_n	$\frac{1}{2^{n/2}\Gamma(n/2)}x^{n/2-1}e^{-x/2}$	$0<x<\infty$, $n>0$ integer	$(1-2\theta)^{-n/2}$	n	$2n$
Beta	$\beta(\alpha_1,\alpha_2)$	$\frac{\Gamma(\alpha_1+\alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)}x^{\alpha_1-1}(1-x)^{\alpha_2-1}$	$0<x<1$, $\alpha_1,\alpha_2>0$	not useful	$\frac{\alpha_1}{\alpha_1+\alpha_2}$	$\frac{\alpha_1\alpha_2}{(\alpha_1+\alpha_2)^2(\alpha_1+\alpha_2+1)}$
Cauchy	–	$\frac{\lambda}{\pi(\lambda^2+(x-\mu)^2)}$	$-\infty<x<\infty$, $\lambda>0$	doesn't exist but char. fn. is $e^{i\mu\theta-\lambda\lvert\theta\rvert}$	no moments exist	
Multivariate Normal	$N(\mu,\Sigma)$	$\frac{\exp\left\{-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right\}}{(2\pi)^{d/2}\det(\Sigma)^{1/2}}$	$x \in \mathbb{R}^d$	$e^{\mu^T\theta+\frac{1}{2}\theta^T\Sigma\theta}$	μ	covariance matrix Σ

Some Discrete Distributions
(All distributions below have integer-valued ranges)

Name	Notation	prob. fn.	range	m.g.f.	mean	variance
Binomial	$\text{Bin}(n,p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	$0\le x\le n$ $0<p<1$	$[(1-p)+pe^\theta]^n$	np	$np(1-p)$
Poisson	$\text{Pois}(\lambda)$	$\frac{e^{-\lambda}\lambda^x}{x!}$	$0\le x<\infty$ $\lambda>0$ real	$e^{\lambda(e^\theta-1)}$	λ	λ
Hyper-geometric	$\text{Hgeo}(n,a,b)$	$\frac{\binom{a}{x}\binom{b}{n-x}}{\binom{a+b}{n}}$	$0\le x\le n$ $a,b>0$ integers	not useful	$\frac{na}{a+b}$	$\frac{nab(a+b-n)}{(a+b)^2(a+b-1)}$
Negative binomial	$\text{Nbin}(k,p)$	$\binom{x-1}{k-1}p^k(1-p)^{x-k}$	$k\le x<\infty$, $0<p<1$	$p^k[e^{-\theta}-(1-p)]^{-k}$	k/p	$k(1-p)/p^2$
Geometric	$\text{Geo}(p)$	$p(1-p)^{x-1}$	$1\le x<\infty$ $0<p<1$	$p[e^{-\theta}-(1-p)]^{-1}$	$1/p$	$(1-p)/p^2$

Multinomial: multivariate extension of binomial

Prob. fn. is given by

$$f(x_1,x_2,\ldots,x_k) = \frac{n!}{x_1!x_2!\ldots x_k!x_{k+1}!}p_1^{x_1}p_2^{x_2}\ldots p_k^{x_k}p_{k+1}^{x_{k+1}}, \quad 0\le x_i\le n, \quad 0<p_i<1 \quad \forall i=1,2,\ldots,n+1$$

where $p_1+p_2+\cdots+p_{k+1}=1$ and $x_1+x_2+\cdots+x_{k+1}=n$.

The m.g.f. is given by

$$[p_1e^{\theta_1}+p_2e^{\theta_2}+\cdots+p_ke^{\theta_k}+p_{k+1}]^n.$$

Moments: $E(X_i)=np_i$, $\text{Var}(X_i)=np_i(1-p_i)$, $\text{cov}(X_i,X_j)=-np_ip_j$, $i,j=1,2,\ldots,n+1$.

1. A travel agent operates a shop where he offers a variety of holiday packages for sale. Below are the prices (in £) for 15 packages sold on a specific day.

99	169	299	350	475
485	550	650	689	830
999	1015	1050	2100	2315

Note that for this sample of data, $\sum x_i = 12,075$ and $\sum (x_i - \bar{x})^2 = 5,779,310$.

(a) Calculate the mean and standard deviation of the holiday packages sold.

[2 marks]

(b) Find the median and the inter-quartile range.

[3 marks]

The travel agent also sells holiday packages online. On the same day, he recorded the price (in £) of 20 packages sold on the website, with the cheapest priced at £98, the most expensive at £2400, and three quartiles at £305, £1380, and £1805. No outliers were observed for sales of holidays sold online.

(c) Draw boxplots for the sales of holiday packages for both in the shop and on the website, using the same scale. Then, comments on the sales data from these two sources.

[4 marks]

(d) Recommend a suitable numerical measure of spread for comparing the dispersion of the two data sets above, and provide your justification without performing any calculations.

[1 mark]

[Total 10 Marks]

2. The lifetime of a certain brand of battery (measured in tens of hours) is modeled by the continuous random variable X with the probability density function given by

$$f_X(x) = \begin{cases} \frac{2}{75}x & 0 \leqslant x \leqslant 5, \\ \frac{2}{15} & 5 < x \leqslant 10, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Explain why f_X is a density function for a random variable X. Give reasons for your answers.

[1 mark]

(b) Determine $\mathrm{Im}X$, the image of the random variable X and find F_X, the cumulative distribution function for X.

[3 marks]

Hence, calculate

(i) $P(X > 4)$ [1 mark]

(ii) the median, m, of the random variable X. [1 mark]

Assuming the lifetimes of the batteries are independent of each other. Two of these batteries are required for an electronic device to operate, and it will function only if **both** batteries are operational.

(c) If two batteries are installed in this device, determine the probability that the device will stop working within the next 40 hours.

[2 marks]

[Total 8 Marks]

3. (a) Let W be a random variable with probability density function given by

$$f_W(w) = \begin{cases} w^{-2} & w > 1, \\ 0 & \text{otherwise.} \end{cases}$$

(i) Find F_W, the distribution function for W. [2 marks]

Suppose that $T = \min\{W_1, W_2, \ldots, W_n\}$, where W_i are independent and identically distributed random variables following the above probability density $f_W(w)$, for $i = 1, \ldots, n$.

(ii) Show that the density function $f_T(t)$ of T is given by

$$f_T(t) = \begin{cases} nt^{-n-1} & t > 1, \\ 0 & \text{otherwise.} \end{cases}$$

[3 marks]

(iii) If $n = 3$, let $T_1 = \min\{W_1, W_2, W_3\}$, find $P(T_1 > 2)$. [2 marks]

(b) The time until a major engine failure in a new car is exponentially distributed with mean of 8 years. If the first major engine failure occurs during the first three years of the car's life, the cost of the repair is covered by the manufacturer's warranty. If the first major failure occurs after three years, the cost to the repair is £ 8000.

(i) Find the probability that first major engine failure in a new car occurs after three years. [2 marks]

(ii) Suppose Y be the cost of the first major repair of a new car's engine that is paid by the car owner. Find $E(Y)$, the expected value of Y. [2 marks]

[Total 11 Marks]

4. The (cumulative) distribution function of a continuous random variable X is given by

$$F_X(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{2}x & 0 \leqslant x \leqslant 1 \\ \frac{1}{4}(1+x) & 1 < x \leqslant 3 \\ 1 & x > 3 \end{cases}$$

(a) Determine ImX, the image of X.

[1 mark]

(b) Does the random variable X have a probability mass function? Give a reason for your answer. [1 mark]

(c) Find:

(i) $P(X > 1)$ [1 mark]

(ii) $P(1 < X \leqslant 2)$ [1 mark]

(ii) $P(X > 1 | X < 2)$ [1 mark]

(d) Find f_X, the probability density function for X.

[1 mark]

(e) Calculate $E(X)$, the expected value of X.

[2 marks]

[Total 8 Marks]

5. Let the joint probability density function of the random variables X and Y be

$$f_{X,Y}(x,y) = \begin{cases} 2(x+y) & 0 < x < 1,\ 0 < y < x, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find the marginal density $f_X(x)$ for X.

[2 marks]

(b) Determine $E(X)$ and $Var(X)$, given that $E(X^2) = \frac{3}{5}$.

[2 marks]

(c) Show that $E(XY) = \frac{1}{3}$.

[2 marks]

(d) Given that $E(Y) = \frac{5}{12}$ and $Var(Y) = \frac{43}{720}$, show that the correlation between X and Y is $\rho(X,Y) = 0.4402$.

[3 marks]

Comment on the correlation between the variables X and Y.

[1 mark]

(e) Show that the conditional density $f_{Y|X}(y \mid x)$ is given by

$$f_{Y|X}(y|x) = \begin{cases} \frac{2}{3}\left(\frac{1}{x} + \frac{y}{x^2}\right), & 0 < y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

[2 marks]

Hence, calculate $P(Y \leqslant \frac{1}{3} | X = \frac{1}{2})$.

[2 marks]

[Total 14 Marks]

6. The discrete random variable X has a probability mass function given by

$$p(x) = \begin{cases} k(2x+3), & x = 1, 2, 3 \\ 0, & \text{otherwise.} \end{cases}$$

(a) Show that $k = \frac{1}{21}$.

[1 mark]

(b) Determine $E(X)$ and $Var(X)$.

[3 marks]

(c) Suppose that $X_1, X_2, \cdots X_{100}$ are independent and identically distributed random variable, each having same distribution as the random variable X of part (a) above and $S = \sum_{i=1}^{100} X_i$.

Use the Central Limit Theorem to approximate $P\,(210 \leqslant S \leqslant 230)$.

[5 marks]

[Total 9 Marks]

END OF PAPER